Institute of Technology of Cambodia
Department of Applied Mathematics and Statistics

Advanced Programming for DS

CHARACTER ERROR RATE (CER)

Lecturer: Mr. Vann Visal

Group 9

PHYRUN Pichchhorda.	e20220895
NOB Sreynich	e20220741
SOUR Alya	e20220969
HOUN Sopanha	e20220209
CHOUB Botumraksa	e20221709
CHHO Sengmeng	e20220296
HEAM Visal	e20221349

INTRODUCTION TO CER

A character-level metric for measuring text differences

WHAT IS CER?

Character Error Rate. It measures how different two texts are, character by character.

CER is the smallest number of character edits from reference to prediction, divided by the reference length.

WHY DOES IT MATTER?

It gives us one simple number for judging speech-to-text, OCR, spelling correction and LLM outputs.

LOWER CER = BETTER

CER focuses on the form of the text (characters), not whether two sentences have the same meaning.

HOW CER WORKS

Compare a model output with a trusted reference

REFERENCE TEXT

The correct text

PREDICTED TEXT

The model output

EDIT DISTANCE

Count S, D and I

CER SCORE

Divide by reference length

Count the fewest single-character edits from reference to prediction, then divide by the reference length.

MATHEMATICAL FORMULA

CER is based on Levenshtein edit distance

$$CER = (S + D + I) / N$$

S — Substitutions

A character is replaced.

a is replaced by u

cat → cut

D — Deletions

A reference character is missing.

l is missing

plane → pane

I — Insertions

An extra character appears.

r is added

cat → cart

N — Reference length

Total characters in the reference.

"chat": N = 4

CER = 0 means an exact match. Multiply by 100 for a percentage: 0.3333 ≈ 33.33%. CER can exceed 100%.

THREE TYPES OF CHARACTER ERRORS

Each edit has a cost of one

SUBSTITUTION

a is replaced by u

cat → cut

DELETION

l is missing

plane → pane

INSERTION

r is added

cat → cart

The total edit distance is the minimum number of substitutions, deletions, and insertions.

PRACTICAL EXAMPLES

Worked calculations using short strings

EXAMPLE 1 — ONE SUBSTITUTION

Reference: cat
Prediction: cut
Edits: S = 1, D = 0, I = 0
Reference length: N = 3
CER = (1 + 0 + 0) / 3 = 0.3333 (33.33%)

EXAMPLE 2 — ONE INSERTION

Reference: cat
Prediction: cart
Edits: S = 0, D = 0, I = 1
Reference length: N = 3
CER = (0 + 0 + 1) / 3 = 0.3333 (33.33%)

Use the fewest edits, not just differences at the same position.
Source: JiWER character processing documentation (linked).

PYTHON IMPLEMENTATION

Dynamic programming saves smaller comparison results to find the fewest edits.

```python
import jiwer

reference = "kitten"
prediction = "sitting"

cer = jiwer.cer(reference, prediction)

print("Reference:", reference)
print("Prediction:", prediction)
print(f"CER: {cer:.2%}")          # CER: 50.00%
```

```
Reference: kitten
Prediction: sitting
CER: 50.00%
```

INPUT

Ref = “Kitten”

Pred = “sitting”

OUTPUT

Return the CER result

EXPERIMENTAL SETUP

A fair comparison requires the same prompts and references

1. SELECT MODELS

Choose two LLMs (Model A and Model B). Record exact model names and versions.

2. PREPARE TEST SET

Use the same 5–10 prompts and write one reference answer for each.

3. CALCULATE CER

Run both outputs through the same Python CER function.

Use the same text rules: state whether spaces, capitals, and punctuation count. Python counts Unicode code points, which may differ from visible symbols. Record model names and test date.
Source: Python Unicode HOWTO (linked).

EXPERIMENTAL RESULTS

Compare CER on the same test items; lower means a closer character match.

Model	Character errors (CER) ↓	Lines fully right ↑
Tesseract 5 (khm+eng, best)	20.55%	16.5%
Netra-OCR	5.87%	56.9%
Chakkhu	**4.50%**	**58.8%**

Chakku has the lowest CER in the results shown.

ADVANTAGES AND LIMITATIONS

CER is useful, but it does not measure meaning

ADVANTAGES

- Simple to explain
- Detects small spelling differences
- Does not need word splitting
- Useful for OCR and speech transcription

LIMITATIONS

- Does not understand meaning
- Can penalize correct rewordings
- Depends on spaces and punctuation
- Character units and text cleanup matter

A low CER does not automatically mean that an answer is factually correct or meaningful.

APPLICATIONS AND CONCLUSION

Use CER when character-level accuracy matters

OCR

Recognizing printed or scanned text

SPEECH

Evaluating automatic speech recognition

SPELLING

Measuring spelling correction errors

LLM OUTPUT

Checking exact names, codes, or fixed-format text

Conclusion: Use CER when exact text matters. For open-ended answers, also check meaning and facts. A close text match alone does not prove that an answer is correct.

THANK YOU